

11. Answer: C

The spacecraft will be attracted towards the Earth due to the gravitational force acting on the spacecraft by the Earth. The spacecraft will either be accelerating towards the Earth or decelerating away from the Earth as shown by **A** and **D** respectively. The gravitational force acting on the spacecraft provides the necessary centripetal force needed by the spacecraft to move in circular path as shown in **B**. When the spacecraft fires its rocket, the spacecraft will move in direction tangent to the path at the position when the rocket is fired as shown in **C**.